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A Startling Tale
of SATURN'S
Radium Madness

"MOMUS MOON"

by Eando Binder

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EDITED BY

ORLIN TREMAINE



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*** START OF THE PROJECT GUTENBERG EBOOK THE ULTIMATE
IMAGE ***

THE ULTIMATE IMAGE

By P. SCHUYLER MILLER

The Magnificent Defense Unit of Dampier.

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"Mike!"

It was Bill Porter's voice. I put one hand on the balustrade and vaulted into the garden. From behind a mass of shrubbery came sounds of a struggle, and Bill's voice rose again.

"Mike, you ape! Step on it!"

I plowed through where someone had gone before. Bill, his shirtfront awry, his coat-tails torn and muddy, was grappling with a snarling, kicking little man about half his size. As I burst out of the shrubbery, Bill kicked his legs from under him and they went down in the newly spaded earth, Bill on top. Bill Porter weighs a good two hundred pounds. The struggle ended then and there.

Bill sat up, one fist clenched in the little man's shirt front. He glared at me out of a rapidly closing eye.

"Where in blue blazes have you been?" he demanded. "D'you think I *like* wrestling with wildcats?"

I looked him over. "Didn't make out so well, did you? Lucky he wasn't any bigger, or I *would* have had to help you. Why pick on a little guy like that? What's he done that you don't like?"

He pointed. Light from the reception hall fell through the bushes in irregular patches. In one of them, half buried in the scuffed-up dirt, I caught the glint of polished metal.

"Pick it up," Bill said.

It was a gun, bigger than the largest six-shooter ever toted by a Hollywood buckaroo. It had a massive stock and the thickest barrel I had ever seen. The whole look of the thing was crazy, like something out of another world.

Bill had been scrambling around in the dirt. I saw that blood was oozing from a gash in his neck. Before I could speak he held up a piece of gleaming metal.

"Take a look at that," he said grimly. "That's what he wanted to pump into the Ambassador. Only I got it instead—in the neck. Now will you give me a

hand with this he-cat before he comes to and starts trying to skin me alive?"

I took the thing. It was a steel bolt or arrow of the kind once used in cross-bows, sharpened to a needle point with six razor-edged vanes running back to the hilt. I slipped it into the chubby muzzle of the gun. It was a perfect fit.

"That," Bill told me, "is a solenoid-gun—one that works. You've seen a metal core pop out of an electric coil when the juice is snapped on. It's a common laboratory stunt. Well, it's grown up and had pups, and this is one of the nastiest of them. No noise at all—and does that dart travel! It would go through a man like cheese even if he's as thick as His Magnificence yonder."

Through the open doors of the reception hall I could see the broad Teutonic back of Herr Wilhelm Friedrich Nebel, Ambassador from the newly stabilized Middle-European Confederacy. Half the stuffed shirts in Washington were crowded around him, trying to make themselves heard over the blare of the band and I recognized three of the President's own private bodyguards. I knew that there were Secret Service men posted all over the grounds to forestall this very thing, yet in spite of them this little man with the outlandish gun had crept within fifty feet of his goal. Had he picked them off, one by one, with his silent darts?

The man was stirring. Bill had him now in a grip that would take more than wildcat tactics to break. I parted the bushes so that a shaft of light fell on his face. Surely I knew that forked beard, those piercing black eyes, the shock of bristling hair. Suddenly I remembered. "Bill! It's Dampier!"

Pierre Dampier, France's greatest physicist, the confrere of Einstein and Heisenberg and Poincare, who had dropped out of sight so mysteriously five years before. Dampier here, in Washington, sniping at the Middle-European Ambassador with an electric gun!

The little man was staring at me with those beady eyes. For a moment I thought he would deny it. Then his face changed. The fury, the madness went out of it and were replaced by a great weariness that made him seem years older. He slumped in Bill's grasp, then stiffened proudly.

"Yes, gentlemen," he admitted. "Pierre Dampier, at your service."

This was no ordinary assassination. Big as the news was, Dampier made it bigger. And news was what Bill and I were here for.

"Bill," I said, "this is our story. No one else even suspects it. Are you going to turn him over to the police or do we get the whole yarn, ourselves, first?"

He nodded. "You're right," he agreed. "We'll never get it if we let him go now. Washington has a way of hushing those things up." He turned to the little Frenchman. "Monsieur Dampier we are newspaper men, we two. There's a reason for what you tried to do tonight, a good reason, or you wouldn't have attempted it. Will you tell us that reason, and let us explain to the world why the great Pierre Dampier has chosen to play the role of a common murderer?"

Dampier stiffened. The forked beard was thrust stiffly forward and the thin shoulders squared in spite of Bill's numbing grip. "I am no murderer!" he hissed. "Wilhelm Nebel is the enemy of my country and of yours—of the world! I stood in his way, and I was crushed. I rose again, and he has found me and tried to grind me under his accursed heel! He will kill me, if I do not kill him first. I implore you, Monsieur, let me go! Let me finish what I have begun. The world will be better for it, and"—a whimsical smile twisted his thin lips—"it will be a greater *coup* for you, will it not?"

Bill was studying him. "We can't do that," he replied, "even if we wanted to. Herr Nebel is our country's guest. But this I will do. Give me your word that you will make no further attempt on Herr Nebel's life for twenty-four hours, tell us why you have done this thing, and I'll let you go. I'll give you one hour's start, and then I'll tell the police the whole story. Is it a bargain?"

Dampier bowed his head. "You have my word, Monsieur. I will tell you everything. But when you have heard what I will say, perhaps you will not wish to call your police. Shall we go to my laboratory? We can talk more freely there."

Bill's grip tightened. "Wait! This garden was guarded. Have you killed those men? Because if you have all bets are off!"

The little Frenchman smiled. "But no, Monsieur. I have no quarrel with your countrymen. There are other missiles for this little toy of mine—hollow needles filled with a certain rare drug like the 'mercy bullets' of your American sportsmen. They will sleep soundly for some hours yet, and have

what you call the big hangover when they awaken but that is all. Shall we go now? It is late, and I have much to tell you."

The whole idea looked screwy to me. Even now I'm not sure that it wasn't. But when Bill Porter makes up his mind, it would take Gabriel's trumpet to change it. He was quite capable of plumping one of Dampier's little needles into me and going off with the Frenchman alone.

"I'll get the car," I said. "Let's get out of here before someone stumbles over a corpse and yells for the cops."

We were somewhere in the middle of Maryland before Bill let me slow down. He must have had a talk with Dampier while I was getting the car, for the little Frenchman never peeped until we swung into a narrow dirt road somewhere north of Frederick. He called the next turn, and the next, until I began to suspect that he was running us around in circles. At last we pulled up before a deserted farm-house, set back from the road behind a dilapidated picket fence. Bill nudged me. Silhouetted against the stars were the towers of a high-tension line. Dampier was either stealing or buying power in a big way.

Now a French gentleman's word is supposed to be about as good as Finland's credit, but we were taking no chances. I remembered that wicked little dart with its razor-edged barbs, and I felt pretty sure that Bill hadn't forgotten it either. We lined up, one on each side of him, and marched across the weed-grown lawn to the rickety side porch. There was a Yale lock on the door, and as Dampier swung it open I saw that it was backed with steel armor-plate. Outside the house might look like the poorer section of Bilded Road, but inside it was built like a fortress. Six-inch concrete walls, steel doors, indirect lighting and ventilation—it looked as though Monsieur Pierre Dampier had been expecting to stand a pretty heavy siege.

A winding stair went down through the floor into a basement room that ran under the entire house. Dampier led the way, Bill followed, and I came last. Probably our science editor could have made something of what Dampier had in that buried room. I couldn't. I wouldn't even have known where to begin photographing it, if the Leica hadn't been back on the terrace at the Embassy where I'd dropped it to vault over the rail into Bill's little shambles, and the Graflex somewhere in the back of the car.

To begin with, he was drawing more current than any ten men I'd ever seen, and I've covered some of the atom-busting at M.I.T. and the lightning shop at Pittsfield. It all went into two huge buss-bars, that ran across to a kind of cage of interlacing copper loops, standing in the center of the room. They were hung from jointed supports that rose above an insulated block or platform of bakelite, with most of the bulkier apparatus inside out of sight, but I had a hunch that whatever was going to happen would take place in, at, and around those spidery coils.

One corner of the room was a kind of office with a desk and books, and a couple of ancient chairs. Dampier waved Bill and me into them and began to pace up and down in front of us like an expectant father. The wild glint had come back into his eyes, but I've seen enough of scientists to know that that isn't necessarily fatal. Most scientists are half nuts anyway. Bill and I never agreed on that point.

You see, before Bill became a demon reporter, he was the white hope of American science. That's how I met him, trying to cover something I couldn't understand and didn't much want to. He fixed my story up for me, and chiseled in on the season's juiciest murder scandal in return. I came down with a bad case of busted cranium, as a result of following his hunches a little too far, and he wrote my scoop for me. After that it stuck. I claimed then they should have made him science editor, but old Medford is our owner's nephew or something, and besides he's pretty good. Anyway, Bill wouldn't take a desk job. It seems he'd always wanted to feel the pulse of Life—

Dampier's English was good. He'd been educated in England and the United States. But when he got excited he fairly surpassed himself and became heart-breakingly colloquial. Where most foreigners would have broken down into their mother-tongue, he relapsed into gutter slang or worse. I've left that out. It doesn't read as well as it sounds, and besides, nice old ladies like to read these magazines. If only they knew the truth—the real inside truth about some of the yarns that have been told in these pages! I've seen the originals—things that a newspaper wouldn't print for fear of being laughed out of a year's circulation—and with proofs! They happen, believe me. Only I'd never been in one before.

Dampier began with true professional dignity. "Gentlemen," he said, "you have treated me honorably. I shall do the same to you. I shall tell you all! When I am finished, judge then if I have done right to assassinate this monster of the devil!

"Monsieur Crandall recognized in me that Pierre Dampier who vanished from the world of science five years ago. It was Wilhelm Nebel who made me to flee like the wild goose. Nebel—the chief of munitions, the millionaire, the so great diplomat, whose hands reach out to every country, regardless of boundaries or the hatred of races. Even in France I was not safe! The finger of Nebel was in the pie of our government. He twisted it—poof! Spies of the police investigate me. They ask questions. They give me the degrees. But I tell them nothing. They can find nothing. It is all here—here in the grey material!" He tapped his bristling skull. "And when they have gone, I take my books, my papers, what money I can get, and take it on the lam to these United States!"

He stopped for breath and glared at us triumphantly. "I scam," he repeated. "I vanish from the sight of men. Here I am Leon the retired hair-dresser, the man with the big radio. Pierre Dampier is forgotten. But not by the accursed Nebel!

"Here in America is a free country where only the dogs, the automobiles, the husbands must have licenses. There are no foolish papers to carry about, no questions to answer to the police. I can hide like a rat in the mousecheese, and be safe. But not from this son-of-an-unpardonableness Nebel! His men are everywhere. He sees everything. Only here I can protect myself. Here I can kill before I am killed!

"But I see in your eye that I am beating about the gas-works, Monsieur. What is it that the old man Dampier has wrested from Nature, that is of so great value to the famous Nebel? What is the secret for which he has lammed himself here to hide like a flea in the chemise of your charming Maryland? Why is he willing to sail down the great river, to fry on the heated seat, so long as Nebel shall die? I will tell you, gentlemen!"

He drew himself up to every inch of his five feet two. He thrust out a pipe-stem arm and pointed an accusing finger at the mechanism that squatted in the middle of the floor.

"There, gentlemen, is the weapon that will make France supreme! The instrument of defense that makes offense impossible! The weapon that will end war!"

We looked at him, and at it, and at each other. It didn't look like the sort of thing you'd lug out on a battlefield to chase the enemy away. It had even less resemblance to the kind of fortress that I'd heard France was building along the Middle-European border. I began to wonder if, after all, that glint in Dampier's eyes was the holy light of pure science.

"What is it?" Bill asked.

The little Frenchman's chest pushed out until his vest-buttons creaked. Then he zipped forward, his rat's eyes darting from side to side, and hissed in our ears:

"It is total reflection!"

That left me cold, but it didn't Bill. I could see that he had a glimmering of an understanding of what went on, but he was puzzled as to the why, what and how. "How d'you mean?" he asked. "We have total internal reflection in prisms. That's no weapon—or defense either, unless you're figuring on Nebel's crowd developing a death-ray or something like that for the next war."

Dampier chuckled. It was about as self-satisfied a chuckle as I've heard. "Death-rays—maybe. I do not care. Bullets, shells, bombs, I tell you nothing, *nothing* can break through the barrier of total reflection! And it is a weapon as well, to turn the enemy's own strength against him."

Bill was sitting up straight in his chair. "Tell me about it," he said softly.

Dampier wriggled and seemed to settle down like a statue on his two spread legs. Only from the waist up was he alive, talking volubly with both hands and that wagging beard.

"It is simple," he explained. "From the beginning of time, what has been the first defense of mankind? It is the wall, the barrier which the enemy cannot climb, cannot break, cannot penetrate with their weapons. A wall of thorns against the beasts of the darkness. A boulder rolled in the mouth of a cave. Walls of sharpened stakes, of earth and stone, of human flesh and blood! Walls of fire laid down by giant guns. Walls of poisonous vapors through

which no living thing can pass. Always a wall, stronger and stronger, but never perfect. I, Pierre Dampier, have made the perfect wall!

"Look, Monsieur—you have spoken of the reflecting prism. All light that falls on it at the proper angle is diverted, turned back. Walls of steel and concrete, such as I have here about me, will repel the bullets of powerful rifles, the shells of small guns, like the little balls of ping-pong. All these things will protect me from the weapons of my enemies—but they are not perfect. They are not total reflection!

"Look you, again. Always there is some ray that will be of the improper angle, the too great or too small wavelength. Always there is some shell that will batter its way through my walls and kill me. But if I can find a mirror that will turn back all rays, a wall from which all projectiles will rebound, a shield against all the many forces of Nature and of man—then, Monsieur, I have the perfect defense and the perfect weapon!

"See this little mirror in my hand. I flash in your eyes a beam of light—so. You are blinded, no? And if this is not light, but a ray of death that you have hurled against my mirror, it kills *you*—is it not so? If it is a bullet that you shoot at me, it recoils and strikes you down. If it is a bomb, it is thrown back into your trenches, to kill your men. If it is a great force of pressure or attraction, it is diverted, reversed, and it strikes at you while I am safe behind my perfect wall."

Bill was on his feet with that mulish look he has when he's sure he's right. "It's impossible!" he snapped. "No metal can reflect all wavelengths. No substance can resist a force greater than those which created it and hold it together. As for magnetism, gravitation, they're space-warp forces. *Things* can't stop them. Sorry we're not in the market for Sunday features today, and I rather doubt that Herr Nebel is. You've got brains—I'll grant you that. You have some energy source in the handle of that little gun of yours that would turn industry up on its tail overnight. I haven't the slightest doubt in the world that you may have blasted the atom wide open and made it sit up and beg. But there's no substance, known or unknown, that will do what you claim, and there never will be. If you have no objections, Monsieur, we will be on our way, and in exactly one hour I will call the police. Au revoir, Monsieur."

Dampier was hopping from one foot to the other like a hen on ice. "No, no, no, Monsieur!" he cried. "You have not heard all! You must lend another ear! There is no substance that will reflect all things; that is true. Only a fool would believe it. But what of a wall that has no substance—that has no existence in what we call reality but that is as fixed and unshakable as the roots of the universe—a wall, a discontinuity of *Space itself*?"

Bill stopped halfway up the stairs. "Say that again," he demanded.

The little Frenchman's hands went winging out in hopeless resignation. "There are no words! One does not explain the theories of Dirac and Schroedinger in words. There are symbols—the logic of symbols—that can be translated at last into reality that men can see, but there are no words for the things that are born and live only here, in the head, in the think-box. It is here, in these symbols, on these sheets of paper. It is there, in that apparatus which you see. But it is not in words."

Bill wasn't being stopped now. He lives words. "You mean," he said, "that you've hit on a condition of Space—maybe a discontinuity of some kind—that has the property of absolute total reflection? It will reflect all radiations one hundred per cent. Any material body will bounce off without making the slightest impression. Every force exerted on it is turned back on itself—even space-forces like gravitation and magnetism. And you can create that condition at will. Is that what you mean?"

Dampier's black eyes fairly spit sparks. "That is it, Monsieur," he cried. "You have said it with a full mouth! My wall, my zone as I have called it, will reflect completely all things, although it is itself a nothing, without existence in our universe. It lives in the symbols of mathematics, and I have just this day completed the apparatus which will give these symbols reality—which will create the zone as I desire it, in any shape or size. I will show you, and you will believe. And then we shall see about Herr Wilhelm Nebel and his makers of wars!"

Bill frowned. "Dampier, give me those equations. I've got to puzzle this thing out for myself, follow your argument through on paper. Is there any place where I can be quiet?"

"But of course, Monsieur. There, in the room for thermal work, everything will be perfectly quiet. Here are the papers, and while you read, I shall show

Monsieur Crandall the working of the works."

But Bill didn't hear that last. The heavy door of the constant temperature room had closed behind him and insulated him from the world.

I couldn't do much but stand and watch Dampier as he bustled about, tuning up his crazy-looking machine. He talked a blue streak as he worked, but most of it went right over my head. I'm no Bill Porter. I did begin to see why Nebel, if he was behind the world's armaments racket as Dampier claimed, might be pretty anxious to get hold of such a thing before the little Frenchman began peddling it to his best customers. In the right hands it might make war very unfashionable.

Imagine an invaded nation squatting down behind a perfectly reflecting wall. They can't see out, but nothing can get in. Enemy shells bounce off into the enemy lines. Death rays flash back into the faces of those who sent them. Radio is garbled by all kinds of curious echoes and reflections, making communication impossible. Electrical and magnetic apparatus would be subject to strange disturbances. And gravitation—how would it affect that? Would every outside object be attracted to the mirror, or would it be repelled by a kind of negative gravity, lifting it into space, to the moon, the planets, to the very stars? I wish now that I'd known at least a fraction of what Bill did, and had been able to read what he read in these few sheets of neatly written paper. I can only guess, from what Dampier said and from what I saw. What his zone really was—what it could do—I do not know.

I tried to pay attention to what he was doing. The real vitals of his apparatus were in the big insulated block. The thousands of amperes he was drawing from the high-tension lines were merely the kicker that kept the real engine turning. Atomic energy, Bill had guessed. Probably he was right.

The loops and coils above the platform determined the shape that the zone would take. According to how they were set, Dampier explained, he could get any geometrically continuous form—a disc, a paraboloid, anything that geometry can describe. What he was going to make was a sphere.

I'm not at all sure that I'm getting the order of things right. I gathered that the zone must be built up and strengthened little by little; first impermeable to the simplest forms of energy, like light and heat, and then to the more and more complex ones, until at some critical point the whole thing became

absolute. The machine that created it had to be outside, otherwise the zone itself would keep any power from getting through. On the other hand, it might be powered by one of those super-batteries that Dampier had in the grip of his solenoid-gun. With a set-up like that, you could dig a hole and pull it in after you, so to speak. What I wondered was how you get out?

I asked Dampier that one. "There would be no way," he told me. "Once the zone is complete, it is unchangeable—absolute. You would be inside, to us here, but I think that to yourself it would seem that it is we who are inside—that you are in a world all of your own, with its own laws, its own science. They can be worked out, these laws. They are in the equations that Monsieur Porter is reading; but they are very strange and complex. In war, a closed zone would be used only as a trap for the enemy."

"Wait a minute," I objected. "You mean to say that once you've made this thing you can't unmake it?"

"That is right," he nodded. "Once the zone is complete it is a bubble—a nothingness—entirely apart from our Space and Time. The forces build up very rapidly, exponentially, but until the very instant of completion, even if it is one little billionth of a second before that moment, the zone will collapse if the power which builds it is shut off. Never in practice would one go so far. Long before it is complete, such a zone will repel all things that can be directed against it, while the balance of power still remains in the hands of him who has created it. To make it—that is nothing. To destroy it is impossible. But to hold it so in the delicate balance between destruction and completion; that is the triumph of Pierre Dampier! I have calculated it all from the equations. See—here at these red lines each needle must stop. If they go beyond—zut! In the space of a thinking the zone is complete! Beyond control!"

He straightened up, his wirey mop of hair bobbing at my shoulder. "Now, please, if you will watch and remember. The loops are set, so, for the sphere—little, like the apple of the eye. Now I press the first switch, and the second and then the others, three, and four, and five. Now I turn the dials, so, a little at a time. A minute now, while the zone builds, and then you will call Monsieur Porter and show him that this is not all sunshine and honeysuckers that he reads."

The big machine began to hum a deep-throated drone that deepened and strengthened until I could feel it shaking the floor under my feet with each colossal pulse of energy. I wondered about the sympathetic vibrations you read about in the Sunday supplements. Might it not shake the walls down around our ears? But Dampier didn't seem worried. And then I forgot it, for a shadow was beginning to form in the space between the coils.

That's all it was at first—a shadow, the size of a big red polished apple. I could hardly be sure it was there, but there was something queer about the way light acted that showed me where it was. Things behind it disappeared, smothered out by something that wasn't really darkness; and then suddenly it began to shine.

You've seen bubbles of air under water, shining like quicksilver. Well, it was like that. It was flawless, without texture, intangible and shimmering. It was not the thing itself we saw, but the things reflected in it—a little, twisted, shining world swimming in the heart of that ball of distorted space. Peering closer, I saw that the coils which shaped it were glowing with an eerie, frosty white light. I stared, fascinated, and by what? By a half-invisible bubble, like an indoor baseball, conjured up by some legerdemain to make fools of us! It was nonsense! I jerked my eyes away—and saw them.

Three men with guns stood on the little stair, watching us. They were gentlemen, polished, clever gentlemen adroit at the art of death. Their guns were of the kind which Middle-Europe gives to its officers, and their faces were Middle-European faces. They were in formal dress, and one of them held his gloves in his left hand.

Dampier had seen them before I, reflected in the shining sphere. He turned, his back against the control-panel, his white teeth gnawing like a rat's at his black beard. The madness was back in his glittering eyes; madness of a trapped beast.

"So!" he whispered. "Now we shall meet."

They came down the stairs, one after the other. How they had cut their way into that Gibraltar of a house I will never know. They may have been working for days and weeks to break through Dampier's defenses. But they were there.

Resistance was futile. Even Dampier realized that. The three guns urged us back against the wall. Deft fingers searched us but found nothing. The three men stepped back to the foot of the little stair, their guns raised, like a firing squad waiting for the signal. And then, above them, I saw the smiling face of Wilhelm Friedrich Nebel, Ambassador from Middle-Europe.

I hadn't believed Dampier's story until then. It was fantastic, this spy business, with a man like Nebel in the villain's role. Things like that don't happen any more. Yet Wilhelm Nebel stood there with a smile on his heavy lips and no smile at all in his pale little eyes. He came down the stairs, treading silently like a cat. He was like a cat in his black and white evening attire, white-bosomed and sleek. He had in his slender fingers a thick golden chain, with a heavy seal of gold made from an ancient coin. A crimson ribbon stretched across his breast like a line of blood.

Satan at the sacrifice! And then the illusion broke.

Those devil fingers went into the pocket of his vest, brought out thick, steel-rimmed spectacles, perched them precariously on the thin-bridged nose. The massive shoulders slouched over, trousers drew tight across his heavy buttocks as he bent and stared into the shining globe. I had never thought of Nebel as fat or gross, in spite of his size, but that single act showed him to me as a Teuton peddler, stooping to finger the weave of some shoddy cloth, to decide how high a price would be safe and how low a one profitable. Satan from his throne! He stood erect again, but his massive face was red with the effort.

Me he ignored. I was nobody. He bowed to Dampier and again I heard the cloth of his breeches creak.

"We meet again, Monsieur."

Dampier answered nothing. He too had his fine tradition of insolence. Nebel's slim hand flicked toward the machine. "This, I presume, is the great weapon that is to be the salvation of *la belle France*. This shining ball that floats in the empty air. Will you show us what it can do?"

The Frenchman's eyes never left Nebel's suave face as he went to the machine. His fingers darted here and there among the dials, tugging and twisting. Above his head the coils stirred in their massive bearings, and within their compass the silver sphere swelled like an inflating balloon to

the size of a man's head—of a basketball—larger and larger while its shimmering surface took on a steely hardness. We seemed to be staring into unfathomable depths, out of which tiny distorted replicas of ourselves peered curiously. I had a feeling that I was two men, one here in this buried room and the other there in that twisted other room, staring inscrutably into my own eyes.

"Stop!" Nebel's voice rapped in my ears. The sphere was huge—ten feet and more in diameter. "It is large enough," he said. "What else will it do?"

I saw Dampier's eyes then. I knew that this time there would be no stopping him. Step by step I withdrew toward the wall. One of the guards saw me and turned his pistol to cover me, but made no other sign.

Dampier answered. "Many things, Monsieur. If you will watch—?" He pulled up his coat-sleeve, baring his scrawny arm, and clambering up on the platform pushed his hand and arm into the shining sphere. I saw the sweat come out on his forehead with the effort. Already the zone was strong. He withdrew his hand and touched the dials of the control-board. Nebel's eyes were watching every move, his hand in the pocket of his coat. Dampier stepped back. "If the gentlemen will shoot? But I warn you—be wary of the ricochet."

Nebel's finger jerked up. "Rudolf!" The youngest of the three men stepped forward and emptied his gun at the shining globe. The first bullet passed through and spanged against the farther wall; the rest glanced whining from its surface and bit ugly scars from the concrete wall beyond. Dampier's eyebrows raised ever so little.

"You have improved the quality of your guns," he commended. "They are more powerful than I had thought."

"Is that all?"

"Is it not enough? What weapon have your thieving swine stolen that will penetrate what you have seen?"

"Is that all?" Nebel's face was purple with rage. They hated each other bitterly, these two, and Dampier had given him not the slightest satisfaction as yet.

The Frenchman shrugged. "It is not complete. Nothing can pass the completed zone, though it is good enough now for anything your blundering fools have invented or will invent. However—"

He turned to the dials. Then suddenly he wheeled. His thin lips were drawn back in a snarl of fury, his eyes were sunken pools of black hate. With a scream he leapt at Nebel's throat.

The first slug caught him in mid-air. The shock dropped him in a crooked heap. Five more bullets smacked into him as he lay there, then Nebel's polished shoe went out and turned him over on his back. He lay there, a bloody froth on his contorted lips, sneering up at the man who had killed him.

For the first time Nebel turned to me. "It was in self defense. You will remember that, Mr. Crandall, if I decide to let you live." He went to the machine, as Dampier had done, and tapped the dials lightly with his long white fingers.

"These red marks—they are, I suppose, the settings with which Monsieur Dampier was working. He would not go beyond, for me. And yet, they are less than halfway to the limit of the dials. What will happen, if I turn them so—a hair beyond?"

His fingers twisted once, twice, and behind us Bill Porter's voice cried out. "Stop, you fool! Stop!"

He stood in the door of the temperature room, the sheaf of Dampier's notes in his hand. Nebel's thin eyebrows went up. "Mr. Porter! I had forgotten you. And why am I a fool?" His fingers spun another of the dials.

"You murdering Teuton fool!" Bill's tone was venomous. "What do you know about science? Your agents bring you this and that. You pay them or kill them, as may be convenient, but what do you know or care about what they have given you, so long as it can be sold at a profit: Mike, come here."

No one moved to stop me. Bill held out the papers, his thumbs marking a certain line. I saw that the margins were filled with his spidery writing.

"Take that top sheet. Now, look at those readings. Has he reached them yet?"

The figures looked familiar. Of course they were the settings at which Dampier had drawn his little red lines.

"He's past them," I cried. "On all but two."

"On all, my friend." Nebel turned again to the dials. "Bluffing does not work in a game for men."

As he moved Bill sprang. Not at Nebel—not at the machine—but at the two great copper bars that came in through the wall. His lean body fell like a stretched spear across them. There was a burst of flame, the stench of burning flesh, but my eyes had left him. For as he leaped Nebel turned the dials.



Sparks crashed in a crescendo as he threw his body across the giant power cables, in a human short-circuit!

A roar of subterranean thunders shook the room. Vast energies poured into the shining zone. It changed. It was a great mirror of utter blackness, its shimmering silver sheen gone leaving a shell of strange transparency out of which creatures of another world leered crookedly at us. And it began to grow!

Momentum carried it. I know that now. The looped coils were swept aside. The apparatus beneath it buckled and split. Beyond it, Nebel's highborn gunmen gaped aghast. They vanished behind its sleek circumference, but Wilhelm Nebel was not of their stupid breed. With a roar he flung his huge body high across the swelling arc of the sphere's circumference. A moment he slithered on its top, sprawled like a toad, his great face crimson—then it crashed him against the ceiling like a toad under a giant's heel. Fragments of concrete began to fall.

I was up the stair, the remaining sheet of Dampier's equations in my hand. I was at the outer door as the walls buckled and fell in ruin. I was running across the littered lawn, staring over my shoulder at the giant silver globe that towered a hundred feet above me. Then it burst!

The force of the explosion hurled me a hundred yards across the fields. I lay gasping in the wet grass, staring glassy-eyed at the column of violet flame that plumed into the sky. I got shakily to my feet and stared into the smoking pit where Dampier's fortress had been. At last I remembered the scrap of crumpled paper in my hand.

The margins of Dampier's paper were full of Bill's penciled notes. At the end he had added five neat equations, and below them the remaining space was filled with his closely written lines.

"These added equations prove Dampier's analysis to be incomplete," he had written. "Such a totally reflecting zone has every characteristic of the closed, intangible boundary of the Einsteinian universe. It may be considered the boundary of such a universe in miniature, containing every

force and body of the greater outside universe which it reflects. Neither is more real, in the physical sense, than the other. There is no way of disproving that we may not in turn be the images of some greater universe than ours, outside of the Einsteinian boundaries of our Space and Time.

"Jeans, and others, have postulated that the size of such a closed universe must depend upon the number of physical particles included in it, and that it will expand, *as our universe is expanding*, until that size is reached. Dampier's closed zone, containing the same number of image-particles as our own outside universe, must expand *to the same size*, and at a vastly greater rate.

"It may be that the cosmic atom, postulated by Abbe Lemaitre, from which our universe was born, was the creation of some Dampier of a super-universe, who failed to check its growth, and that its swelling bubble is crushing the mighty cosmos of which it is the ultimate image, as Dampier's completed zone would crush our own."

Bill Porter's scribbled notes stop there. In the split millionth of a second before the twist of Nebel's fingers could throw the balanced sphere over the boundary to completion, his body shorted the power that fed the great machine. It was in time! Momentum of growth, gained in that instant of which Dampier had told me, swept Nebel and his gunmen to their death, and as the zone collapsed the incalculable energies trapped in it burst forth in a holocaust of atomic flame. A millionth of a second—less perhaps—but in it chance, and whatever power it is that rules chance, had checked the thing whose illimitable growth would have swept our universe before it in an avalanche of destruction.

If, as Bill Porter thought, our universe is just such a swelling bubble in the vaster world which it mirrors, I wonder whether in that world there is not another Dampier, another Nebel, another Bill Porter going to his death. I wonder if Time itself is not reflected in some contorted scale in such a cosmic bubble, and the entire history of a universe reproduced in the instant before it bursts.

I wonder, too, if one day our bubble-universe will not burst as Dampier's did, robbing us in that future instant of all reality—the snuffed out images in an almost perfect mirror. For as our Dampier did, so did the greater Dampier whose image he was. As he failed so did that other Dampier fail.

Perhaps, in his turn, he but mirrored greater things beyond. Where then—in what inconceivable realm beyond Space and Time—is the reality of which we are the ultimate image?

*** END OF THE PROJECT GUTENBERG EBOOK THE ULTIMATE
IMAGE ***

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